THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS VIII — SCIENCE · MID-TERM PRACTICE PAPER

Session 2026-27 · Term I · Chapters 1–7

Class: 8	Subject: Science	Set 03	Time: 2 Hours 30 Minutes	Max. Marks: 60

General Instructions:

1. All questions are compulsory.
2. This question paper has five sections A, B, C, D and E. Marks are indicated against each question.
3. Section A (Q. 1, parts I–XII) consists of multiple choice questions of 1 mark each.
4. Section B (Q. 2–5) consists of very short answer questions of 2 marks each.
5. Section C (Q. 6–8) consists of short answer questions of 3 marks each.
6. Section D (Q. 9–11) consists of long answer questions of 5 marks each.
7. Section E (Q. 12–15) consists of four case-based / source-based questions of 4 marks each (1 + 1 + 2).
8. There is no overall choice. However, internal choice is given in some questions. Only one of the choices in such questions is to be attempted.
9. Draw neat and labelled diagrams wherever required.

SECTION A — Multiple Choice Questions — 1 questions · 12 marks

Question 1 has twelve parts (I–XII). Each carries 1 mark.

1. From Question I–XII, choose the correct option.

I. A student claims that plants grow faster in music. To test it fairly she should: 1

(a) keep both plants in the dark
(b) use plants of different species
(c) play music to only one plant and give it extra water
(d) grow two identical plants under the same conditions, one with music and one without

II. Which of the following is used to make bread soft and spongy? 1

(a) Lactobacillus (b) Yeast (c) Rhizobium (d) Amoeba

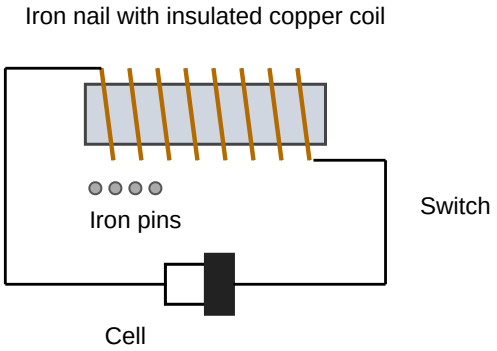
III. A person has repeated cold, fever and is infected by a virus. Which medicine will NOT help in killing the virus? 1

(a) Vaccine given earlier for prevention (b) Rest and fluids
(c) Antiviral drug prescribed by a doctor (d) Antibiotic

IV. A cell in a circuit is shown by a long line and a short line. The longer line represents the: 1

(a) bulb (b) negative terminal
(c) positive terminal (d) switch

V. In the figure of the electromagnet, the iron pins are attracted when the switch is: 1



(a) connected to a wooden nail (b) removed from the circuit
(c) open (d) closed

VI A boy pushes a heavy wall but it does not move. The reason is that: 1

(a) he applied no force

(b) walls cannot experience force

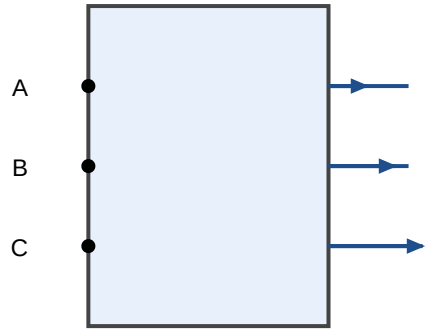
(c) gravity cancels friction only

(d) the net force on the wall is zero as friction and other forces balance his push

VII An object weighs 60 N on Earth. Its mass is nearly (g = 10 m/s²): 1

(a) 0.6 kg (b) 600 kg (c) 60 kg (d) 6 kg

VIII In the figure, water comes out fastest and farthest from hole: 1



Water-filled container with holes A, B, C

(a) All holes equally (b) C, the lowest hole

(c) B, the middle hole (d) A, the topmost hole

IX Which of these will be helpful when a cyclone warning is announced? 1

(a) Fly a kite

(b) Stand under tall trees

(c) Move to a cyclone shelter and keep a radio/battery and emergency kit ready

(d) Go to the seashore to see it

X The pressure exerted by a gas in a closed container is due to: 1

(a) continuous collisions of its particles with the walls

(b) attraction between the particles

(c) gravity pulling the gas down only

(d) its colour

XI Which statement is correct about solids? 1

(a) Particles vibrate about fixed positions (b) Particles move freely everywhere

(c) They have no definite volume (d) There is no force between particles

XII A pressure cooker cooks food faster mainly because: 1

(a) it makes water colder

(b) it removes all air pressure

(c) it uses more water

(d) the increased pressure raises the boiling point of water

SECTION B — Very Short Answer Questions — 4 questions · 8 marks

2 marks each (about 30–40 words).

2. Why is it important to record observations carefully during an investigation? **2**

3. Name two harmful effects of microorganisms. **2**

OR

How do Lactobacillus bacteria help in the formation of curd? (Answer in two points.)

4. Why do electric wires used in houses have a plastic/rubber coating? **2**

5. Why does an inflated balloon burst when it is squeezed hard at a point, but not when pressed by a flat hand? **2**

SECTION C — Short Answer Questions — 3 questions · 9 marks

3 marks each (about 60 words).

6. Differentiate between communicable and non-communicable diseases (any two points) and give one example of each. **3**

OR

List three ways to prevent infectious diseases.

7. An electric heater of a room warms up in a few minutes, yet the wire joining it to the plug stays cool. Explain. **3**

8. A force of 400 N acts on a surface of area 0.2 m². Calculate the pressure. What will be the pressure if the area is doubled for the same force? **3**

OR

Explain why it is easier to cut with a sharp knife than a blunt knife and why heavy vehicles have wide tyres. Use the idea of pressure.

SECTION D — Long Answer Questions — 3 questions · 15 marks

5 marks each.

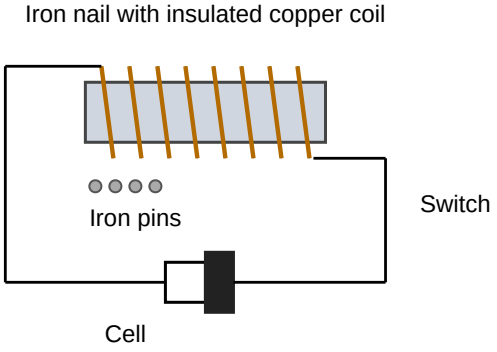
9. Complete the given sentences (choose from the words in the box): **(Antibiotic, Barometer, Hypothesis, Diffusion, Gravity)** **5**

i. A _____ is a tentative explanation that can be tested by experiment.
ii. The spreading of particles of one substance into another on their own is called _____.
iii. A _____ is an instrument used to measure atmospheric pressure.
iv. An _____ such as penicillin kills bacteria.
v. The force of _____ pulls objects towards the centre of the Earth.

10. Match the Columns **5**

Column I	Column II
(a) Amoeba	(i) protozoan
(b) Rhizobium	(ii) fixes nitrogen in root nodules
(c) Nichrome	(iii) alloy used in heating elements
(d) Friction	(iv) force opposing relative motion
(e) Liquid	(v) takes the shape of the container but has fixed volume
	(vi) SI unit of pressure

11. (a) In the circuit shown, a coil is wound round an iron nail. Name the device and state two factors on which its strength depends. (3) **5**
(b) How can its poles be reversed? (2)



OR

(a) Name the three states of matter and give one example of each. (3)
(b) Why can a gas fill the whole container but a solid cannot? (2)

SECTION E — Case-based / Source-based Questions — 4 questions · 16 marks

Read the cases carefully and answer. Each question carries 4 marks.

12. Ravi saw white fur-like growth on a piece of bread kept in a warm damp cupboard for a week.

(a) Which group of microorganisms has grown on the bread? 1
(b) Name one condition that helped the growth. 1
(c) Suggest two ways to prevent food from such spoilage. 2

OR

(d) Name one useful fungus and its use.

13. Sara tested two bulbs: one bulb had a thin tungsten filament and the other a thick copper wire in place of the filament. The tungsten bulb glowed bright, the copper wire only became slightly warm.

(a) What effect of current makes the filament glow? 1
(b) Which has more resistance – tungsten filament or copper wire? 1
(c) Why is tungsten used as the filament material? 2

OR

(d) Name two appliances that work on the heating effect of current.

14. A girl of mass 40 kg stands on a bathroom weighing scale. The scale reads 400 N. The next day her brother of mass 30 kg stands on the same scale.

(a) What quantity does the scale actually measure here? 1
(b) What will the scale read for her brother (g = 10 m/s²)? 1
(c) Would their masses change on the Moon? Would their weights change? Explain. 2

OR

(d) Give two differences between mass and weight.

15. A balloon filled with air is tied tightly and kept in the hot sun. After some time it bursts. Rohit recalled that pressure of a gas depends on the movement of its particles.

(a) What causes the pressure of a gas in a balloon? 1
(b) What happens to the speed of particles when the gas is heated? 1
(c) Explain why the balloon burst in the sun. 2

OR

(d) Why does a gas fill a container completely?